ITI – RedHat Admin 2 lab2

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1. **What Different Between Rsyslog and Journald?**
   Both are logging service, Journal logs as bytes, and it is part of systemd. Rsyslog logs as Text.

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2. **What are the main configuration files for Rsyslog?**

   /etc/rsyslog.conf and custom configuration files.

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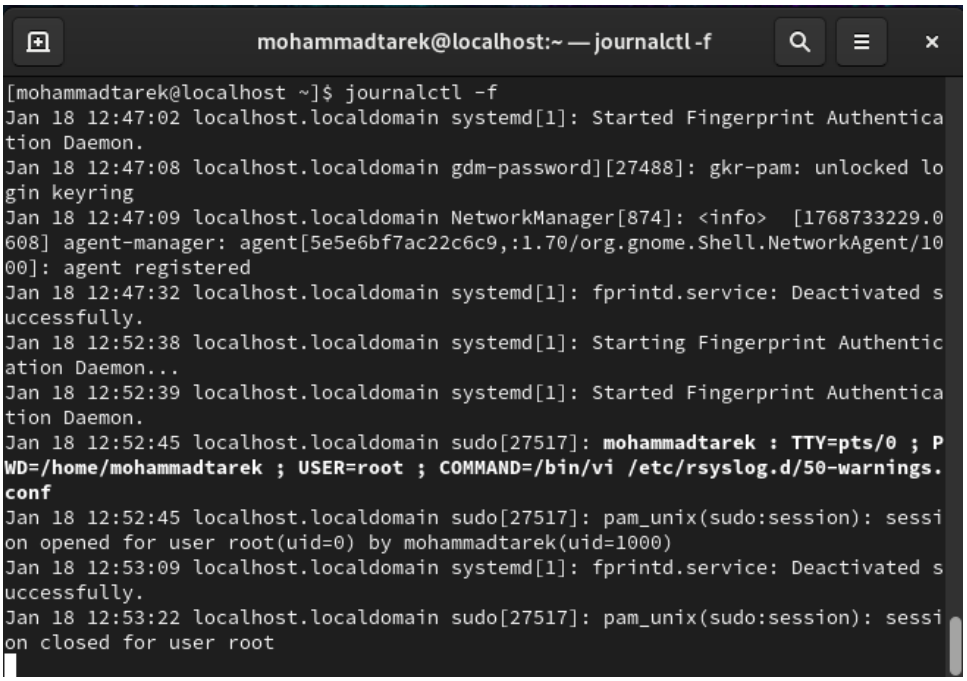
3. **How do you view system logs in real time?**



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4. **How do you test if Rsyslog is working properly after making changes?**

mohammadtarek@localhost:~ — sudo systemctl status rsyslog

mohammadtarek@localhost:~ — ... | mohammadtarek@localhost:~

```
[mohammadtarek@localhost ~]$ rsyslogd -N1
rsyslogd: version 8.2510.0-2.el9, config validation run (level 1), master config
 /etc/rsyslog.conf
rsyslogd: End of config validation run. Bye.
[mohammadtarek@localhost ~]$ sudo systemctl restart rsyslog
[sudo] password for mohammadtarek:
[mohammadtarek@localhost ~]$ sudo systemctl status rsyslog
● rsyslog.service - System Logging Service
     Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: >
     Active: active (running) since Sun 2026-01-18 13:12:24 EET; 26s ago
       Docs: man:rsyslogd(8)
             https://www.rsyslog.com/doc/
   Main PID: 27642 (rsyslogd)
      Tasks: 3 (limit: 10519)
     Memory: 1.1M (peak: 1.4M)
        CPU: 51ms
     CGroup: /system.slice/rsyslog.service
             └─27642 /usr/sbin/rsyslogd -n

Jan 18 13:12:24 localhost.localdomain systemd[1]: Starting System Logging Servi>
Jan 18 13:12:24 localhost.localdomain systemd[1]: Started System Logging Servic>
Jan 18 13:12:24 localhost.localdomain rsyslogd[27642]: [origin software="rsyslo>
Jan 18 13:12:24 localhost.localdomain rsyslogd[27642]: imjournal: journal files>
lines 1-16/16 (END)
```

```
[mohammadtarek@localhost ~]$ logger -p user.info "I am trying to love Linux"
```

```
mohammadtarek@localhost:~ — sudo tail -f /var/log/messages
mohammadtarek@localhost:~        mohammadtarek@localhost:~ — ...
[mohammadtarek@localhost ~]$ sudo tail -f /var/log/messages
Jan 18 13:12:24 localhost systemd[1]: Started System Logging Service.
Jan 18 13:12:24 localhost rsyslogd[27642]: [origin software="rsyslogd" swVersion
="8.2510.0-2.el9" x-pid="27642" x-info="https://www.rsyslog.com"] start
Jan 18 13:12:24 localhost rsyslogd[27642]: imjournal: journal files changed, rel
oading...  [v8.2510.0-2.el9 try https://www.rsyslog.com/e/0 ]
Jan 18 13:12:25 localhost systemd[1]: fprintd.service: Deactivated successfully.
Jan 18 13:14:48 localhost cupsd[888]: REQUEST localhost - - "POST / HTTP/1.1" 20
0 192 Renew-Subscription successful-ok
Jan 18 13:15:19 localhost systemd[1]: Starting Fingerprint Authentication Daemon
...
Jan 18 13:15:19 localhost systemd[1]: Started Fingerprint Authentication Daemon.
Jan 18 13:15:29 localhost mohammadtarek[27690]: I am trying to love Linux
Jan 18 13:15:49 localhost systemd[1]: fprintd.service: Deactivated successfully.
Jan 18 13:16:56 localhost mohammadtarek[27704]: I am trying to love Linux
```

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5. **You need to configure Rsyslog to log messages from any facility with severity warning and above to a file located at /var/log/warnings.log**

```
[mohammadtarek@localhost ~]$ sudo vim /etc/rsyslog.conf
[sudo] password for mohammadtarek:
```

```
mohammadtarek@localhost:~ — sudo vim /etc/rsyslog.conf
mohammadtarek@localhost:~ — ...   mohammadtarek@localhost:~ — ...

# Everybody gets emergency messages
*.emerg action(type="omusrmsg" users="*")

# Save news errors of level crit and higher in a special file.
uucp,news.crit action(type="omfile" file="/var/log/spooler")

# Save boot messages also to boot.log
local7.* action(type="omfile" file="/var/log/boot.log")


# ### sample forwarding rule ###
#action(type="omfwd"
# # An on-disk queue is created for this action. If the remote host is
# # down, messages are spooled to disk and sent when it is up again.
#queue.filename="fwdRule1"       # unique name prefix for spool files
#queue.maxdiskspace="1g"         # 1gb space limit (use as much as possible)
#queue.saveonshutdown="on"       # save messages to disk on shutdown
#queue.type="LinkedList"         # run asynchronously
#action.resumeRetryCount="-1"    # infinite retries if host is down
# # Remote Logging (we use TCP for reliable delivery)
# # remote_host is: name/ip, e.g. 192.168.0.1, port optional e.g. 10514
#Target="remote_host" Port="XXX" Protocol="tcp")
*.warn  /var/log/warnings.log
"/etc/rsyslog.conf" 82L, 3329B                          82,28-29      Bot
```

```
[mohammadtarek@localhost ~]$ sudo touch /var/log/warnings.log
[mohammadtarek@localhost ~]$ sudo chmod x+u /var/log/warnings.log
chmod: invalid mode: 'x+u'
Try 'chmod --help' for more information.
[mohammadtarek@localhost ~]$ sudo chmod U+X /var/log/warnings.log
chmod: invalid mode: 'U+X'
Try 'chmod --help' for more information.
[mohammadtarek@localhost ~]$ sudo chmod u+x /var/log/warnings.log
[mohammadtarek@localhost ~]$ logger -p user.warn "Hello from Linux"
```

```
[mohammadtarek@localhost ~]$ sudo tail -f /var/log/warnings.log
Jan 18 13:28:07 localhost mohammadtarek[27864]: Hello from Linux
```

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6. How can you configure Rsyslog to discard log messages discard logs from a specific facility (e.g., auth)

```
mohammadtarek@localhost:~ — sudo vim /etc/rsyslog.conf
mohammadtarek@localhost:~ — ...   mohammadtarek@localhost:~ — ...
auth.*  stop
```

```
[mohammadtarek@localhost ~]$ sudo vim /etc/rsyslog.conf
[mohammadtarek@localhost ~]$ sudo systemctl restart rsyslog
[mohammadtarek@localhost ~]$ logger -p auth.info
^C
[mohammadtarek@localhost ~]$ logger -p auth.info "Hello again from Linux"
[mohammadtarek@localhost ~]$ logger -p auth.info "Hello again from Linux"
```

```
mohammadtarek@localhost:~   mohammadtarek@localhost:~ — ...
Jan 18 13:31:21 localhost sudo[27870]: pam_unix(sudo:session): session opened fo
r user root(uid=0) by mohammadtarek(uid=1000)
Jan 18 13:32:47 localhost sudo[27870]: pam_unix(sudo:session): session closed fo
r user root
Jan 18 13:33:13 localhost sudo[27879]: mohammadtarek : TTY=pts/0 ; PWD=/home/moh
ammadtarek ; USER=root ; COMMAND=/bin/systemctl restart rsyslog
Jan 18 13:33:13 localhost sudo[27879]: pam_unix(sudo:session): session opened fo
r user root(uid=0) by mohammadtarek(uid=1000)
Jan 18 13:33:13 localhost sudo[27879]: pam_unix(sudo:session): session closed fo
r user root
Jan 18 13:33:55 localhost sudo[27833]: pam_unix(sudo:session): session closed fo
r user root
Jan 18 13:34:11 localhost sudo[27906]: mohammadtarek : TTY=pts/1 ; PWD=/home/moh
ammadtarek ; USER=root ; COMMAND=/bin/tail -f /var/log/secure
Jan 18 13:34:11 localhost sudo[27906]: pam_unix(sudo:session): session opened fo
r user root(uid=0) by mohammadtarek(uid=1000)
Jan 18 13:34:56 localhost sudo[27923]: mohammadtarek : TTY=pts/0 ; PWD=/home/moh
ammadtarek ; USER=root ; COMMAND=/bin/systemctl status rsyslog
Jan 18 13:34:56 localhost sudo[27923]: pam_unix(sudo:session): session opened fo
r user root(uid=0) by mohammadtarek(uid=1000)
Jan 18 13:34:58 localhost sudo[27923]: pam_unix(sudo:session): session closed fo
r user root
Jan 18 13:49:15 localhost gdm-password][27952]: gkr-pam: unlocked login keyring
```

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7. How do you configure Rsyslog to log messages from a specific application to a custom log file?

```
mohammadtarek@localhost:~ — sudo vim /etc/rsyslog.conf
mohammadtarek@localhost:~ — ...    mohammadtarek@localhost:~
if ($programname == "apptest") then /var/log/apptest & stop
```

```
mohammadtarek@localhost:~
mohammadtarek@localhost:~    mohammadtarek@lo
[mohammadtarek@localhost ~]$ sudo vim /etc/rsyslog.conf
[mohammadtarek@localhost ~]$ sudo touch /var/log/apptest
[mohammadtarek@localhost ~]$ sudo chmod u+x /var/log/apptest
[mohammadtarek@localhost ~]$ sudo systemctl restart rsyslog
[mohammadtarek@localhost ~]$ tail -f /var/log/apptest
^C
[mohammadtarek@localhost ~]$ sudo logger -t apptest "test"
[mohammadtarek@localhost ~]$ tail -f /var/log/apptest
```

```
mohammadtarek@localhost:~ — sudo tail -f /var/log/apptest
mohammadtarek@localhost:~    mohammadtarek@localhost:~ — ...
[mohammadtarek@localhost ~]$ sudo tail -f /var/log/apptest
[sudo] password for mohammadtarek:
^C
[mohammadtarek@localhost ~]$ sudo tail -f /var/log/apptest
Jan 18 15:10:39 localhost apptest[29175]: Test
```

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8. Create Your Customer Service

```
mohammadtarek@localhost:~
mohammadtarek@localhost:~    mohammadtarek@localhost:~ — ...
[mohammadtarek@localhost ~]$ sudo vim /usr/bin/admin2lab2.sh
[mohammadtarek@localhost ~]$ sudo chmod u+x /usr/bin/admin2lab2.sh
```

mohammadtarek@localhost:~ — sudo vim /usr/bin/admi

mohammadtarek@localhost:~ — ... | mohammadtarek@

```
#!usr/bin/bash
while true;
do

        echo "This is the last hello for today"
        sleep 5
done
~
~
```

mohammadtarek@localhost:~

mohammadtarek@localhost:~ | mohammadtarek@localhost:~ — ...

```
[mohammadtarek@localhost ~]$ sudo vim /usr/bin/admin2lab2.sh
[mohammadtarek@localhost ~]$ sudo chmod u+x /usr/bin/admin2lab2.sh
[mohammadtarek@localhost ~]$ sudo useradd -M -r -s /bin/false admin2user
[mohammadtarek@localhost ~]$ sudo chown admin2user:admin2user /usr/bin/admin2lab
2.sh
```

mohammadtarek@localhost:~ — sudo vim /etc/systemd/syste...

mohammadtarek@localhost:~ — ... | mohammadtarek@localhost:~ — ...

```
[Unit]
Description=Admin2Lab2 service
After=network.target

[Service]
User=admin2user
Group=admin2user
Type=simple
ExecStart=/usr/bin/bash /usr/bin/admin2lab2.sh
Restart=on-failure
RestartSec=30

[Install]
WantedBy=multi-user.target
~
```

```
[mohammadtarek@localhost ~]$ sudo systemctl restart admin2lab2.service
Warning: The unit file, source configuration file or drop-ins of admin2lab2.serv
ice changed on disk. Run 'systemctl daemon-reload' to reload units.
[mohammadtarek@localhost ~]$ sudo systemctl status admin2lab2.service
Warning: The unit file, source configuration file or drop-ins of admin2lab2.ser>
● admin2lab2.service - Admin2Lab2 service
     Loaded: loaded (/etc/systemd/system/admin2lab2.service; enabled; preset: d>
     Active: active (running) since Sun 2026-01-18 15:38:33 EET; 33s ago
   Main PID: 29561 (bash)
      Tasks: 2 (limit: 10519)
     Memory: 612.0K (peak: 844.0K)
        CPU: 58ms
     CGroup: /system.slice/admin2lab2.service
             ├─29561 /usr/bin/bash /usr/bin/admin2lab2.sh
             └─29573 sleep 5

Jan 18 15:38:33 localhost.localdomain systemd[1]: Started Admin2Lab2 service.
Jan 18 15:38:33 localhost.localdomain bash[29561]: This is the last hello for t>
Jan 18 15:38:38 localhost.localdomain bash[29561]: This is the last hello for t>
Jan 18 15:38:43 localhost.localdomain bash[29561]: This is the last hello for t>
Jan 18 15:38:48 localhost.localdomain bash[29561]: This is the last hello for t>
Jan 18 15:38:53 localhost.localdomain bash[29561]: This is the last hello for t>
Jan 18 15:38:58 localhost.localdomain bash[29561]: This is the last hello for t>
Jan 18 15:39:03 localhost.localdomain bash[29561]: This is the last hello for t>
lines 1-20/20 (END)
```